

Jack Geary

Machine Learning PhD Student

📍 Edinburgh, United Kingdom

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Seeking machine learning research internships | available 2026–2027

Profile

Machine learning PhD researcher specialising in strategic classification and robustness to distribution shift. Three years as an industrial ML/NLP engineer before starting my PhD give me equal footing in research and production.

Research Experience

Postgraduate Researcher

Jun. 2024 – Present

University of Edinburgh, Edinburgh, UK

- Built a publicly available [framework](#) for training and evaluating classifiers in strategic settings, making the research easier to reproduce and extend.
- Developed approximation methods for strategic responses and validated them on real datasets, with the goal of making non-linear strategic classification more practical.
- Proposed methods for training non-linear and randomized classifiers under strategic behavior, and derived conditions under which randomized classifiers are optimal.

Postgraduate Researcher

Sept. 2017 – May 2021

University of Edinburgh, Edinburgh, UK

- Published work on game-theoretic decision-making in autonomous driving, with an emphasis on modeling interaction rather than treating agents independently.
- Developed active parameter inference methods for settings with unknown agents, using exploration to learn latent behavioral parameters.
- Designed and implemented simulation tooling for driving experiments, supporting empirical evaluation of interactive decision-making methods.

Industry Experience

Data Scientist

Jun. 2021 – Jun. 2024

Ripjar, Edinburgh, UK

- Worked on machine-learning pipelines for extracting and processing entity information from unstructured text in a financial-crime context.
- Designed system and statistical tests to track model quality, catch regressions early, and make model changes safer to ship.
- Implemented and maintained entity-linking algorithms on large datasets, balancing model quality, robustness, and operational reliability.
- Chaired the company's Machine Learning Reading Group and represented the company at international academic conferences.

Education

Ph.D. Informatics

2024 – Present

University of Edinburgh, Edinburgh, UK

M.Phil. Informatics

2017 – 2021

University of Edinburgh, Edinburgh, UK

M.Sc. Artificial Intelligence

2016 – 2017

University of Edinburgh, Edinburgh, UK First Class Honours

B.A. Mathematics

2012 – 2016

Trinity College Dublin, Dublin, Ireland First Class Honours; Gold Medal for academic excellence

Selected Publications

Non-Linear Strategic Classification Made Practical

Jack Geary, Boyan Gao, Henry Gouk

Under review, NeurIPS

2026

Approximates best responses for non-linear classifiers in strategic settings via Lagrangian duality, enabling strategic training and evaluation beyond the linear case. Extends and supersedes an earlier workshop version presented at ICML 2026 (NExT-Game).

The Signal in the Noise: OOD Detection Through Goodness-of-Fit Testing in Factorised Latent Spaces

Philipp Bomatter, Jack Geary, Henry Gouk

Pre-print

2026

Uses the mass-preserving properties of continuous normalizing flows to flag out-of-distribution samples via goodness-of-fit testing in latent space, addressing the unreliability of generative-model likelihoods for OOD detection.

Strategic Classification with Randomised Classifiers

Jack Geary, Henry Gouk

Pre-print

2025

Develops randomized classifiers for strategic settings and provides theory showing when they can outperform deterministic models.

Active Altruism Learning and Information Sufficiency for Autonomous Driving

Jack Geary, Henry Gouk, Subramanian Ramamoorthy

Pre-print

2021

Uses active learning for parameter inference in game-theoretic autonomous-driving settings with unknown agents.

Resolving Conflict in Decision-Making for Autonomous Driving

Jack Geary, Subramanian Ramamoorthy, Henry Gouk

Robotics: Science and Systems 2021

2021

Reports human-subject evidence and a revised altruism formulation that better matches behavior in interactive driving scenarios.

Altruistic Decision-Making for Autonomous Driving with Sparse Rewards

Jack Geary, Henry Gouk

Robotics: Science and Systems 2020

2020

Shows how altruism-aware decision-making can reduce equilibrium mismatch in Stackelberg-game-style driving scenarios.

Skills

- **Languages:** Python, C/C++ , Prolog
- **ML frameworks & tools:** PyTorch, scikit-learn, Pandas, PySpark, Git
- **Machine learning:** deep learning, theoretical and empirical ML, strategic classification, game theory
- **Practices:** reproducible research, academic writing, $\text{\LaTeX}$